From Aggregate Scores to Warranted Inference in AGI Evaluation

Brett Reynolds [*]
Humber Polytechnic & University of Toronto

Abstract

Test batteries spanning several cognitive domains are now offered as measures of progress toward artificial general intelligence, and their scores read as evidence about tasks, contexts, systems, and times that were never tested. Precision within a standardized setting doesn't license that reading. Standardization narrows what a score represents even while stabilizing it, and aggregation discards distinctions within what remains. A stable average is consistent with unchanged items, with gains and losses that cancel, and with persistent severe failures.

Validity theory locates warrant in the argument for a proposed interpretation and use, not in the test alone. I use PROJECTIBILITY for the evidential standing of one specified source–target link: a claim extending an interpretation, prediction, or explanation beyond the observations that produced the score. Goodman's grue case shows why fit to those observations can't settle which extension is supported. Evaluators should declare each link, preserve the distinctions it makes relevant, and test it with evidence matched to the facet it crosses.

A reanalysis of 32 released model–benchmark comparisons finds a small average decline coexisting with a much larger change tail, and simulations with known latent truths find a zero change tail coexisting with a high absolute risk tail. Both show that raw, null-referenced, and split-sample estimators target different objects. Neither shows that any quantity predicts an untested target. No developer can anticipate every downstream use, so evidential labour divides: developers characterize bounded source evaluations and expose unsupported facets; deployers complete and monitor local warrants. Bounded successes don't compose into unrestricted generality.

Keywords: AGI evaluation; validity; projectibility; tail risk; generalization

1 The Inferential Overreach of AGI Scores

Multidomain evaluations are often invoked to support inferences or decisions beyond their observed results. Aggregation necessarily compresses. The problem is that evaluators rarely establish whether the omitted distinctions matter to the inference or decision.

[*] I used ChatGPT 5, Claude Sonnet 4.5, and Gemini Pro 2.5 extensively in drafting and revising this paper. I reviewed, edited, and approved all the material and take full responsibility for the final text and conclusions. brett.reynolds@humber.ca

This paper adapts validity theory from educational and psychological measurement. Its central move is to treat validity not as a property of a test in isolation but as the strength of the argument for a proposed interpretation of its scores and use of them (Kane, 2013; Messick, 1995b). A score may be precise and repeatable for a restricted set of observations while providing weak evidence for a broader claim.

Goodman's new riddle of induction isolates the underlying problem. Suppose that every emerald examined before a specified time t has been green. Those observations fit the familiar hypothesis that all emeralds are green.

Now define a second predicate. An object is *grue* if it was examined before t and is green, or if it wasn't so examined and is blue. The same observations fit the hypothesis that all emeralds are grue. The two hypotheses agree about every emerald observed so far but disagree about an emerald first examined after t: the green hypothesis predicts a green emerald; the grue hypothesis predicts a blue one. The riddle asks why the first is PROJECTIBLE, or fit for projection beyond the observed cases, while the second isn't (Goodman, 1955/1983, pp. 74–80).

Goodman answers the riddle by appealing to the history of inductive practice, not to a better fit with the observed emeralds: green and grue fit them equally well. *Green* has repeatedly figured in past projections; *grue* hasn't. He calls this history ENTRENCHMENT and uses it to distinguish otherwise equally supported predicates (Goodman, 1955/1983, pp. 84–86, 93–98). Boyd (1991, pp. 136–141) places more weight on empirically revisable background theory and, for inductive or explanatory categories, causal structure. I borrow the question and vocabulary of projectibility without treating either entrenchment or causal structure as a sufficient criterion of warrant.

The analogous question for AI benchmarks is whether success on scored items warrants a broader claim. Observed successes may fit both the claim that a system performs well across a domain and the rival claim that it performs well only on tasks sharing the tested formats or contexts. The claims agree on the scored items and diverge on the cases that motivate broader evaluation. I adapt PROJECTIBILITY to name the evidential standing of a specified source–target link that extends an interpretation, prediction, or explanation beyond the observed evaluation.

A running example makes the issue concrete. A law firm is deciding whether to release a legal-research assistant built on a model whose developer publishes strong scores across a multidomain battery and describes it as broadly capable. The firm needs to know what those scores tell it about its own next model–retrieval–tool workflow, on local matters, when used by its lawyers, with clients and courts bearing the consequences. The published battery is the source; the firm's later workflow and matters are the target.

The source score comes from a standardized evaluation, which fixes prompts, scoring, and conditions so that repeated evaluation answers a stable question. Standardization also narrows the EVALUATION UNIVERSE: the tasks, prompts, scoring rules, and conditions represented by the score.

Kane distinguishes two moves: generalizing from observed scores to a test's standardized universe and extrapolating from that universe to a broader target domain (Kane, 2013, pp. 22–25). Figure 1 places those moves in the longer chain the running example needs. Brennan shows why the distinction matters: a feature held fixed in the restricted universe may vary in the broader one, so standardization can raise precision within the former while weakening extrapolation to the latter (Brennan, 2001, pp. 132–135). More observations from the narrowed universe leave systematic underrepresentation of the target intact.

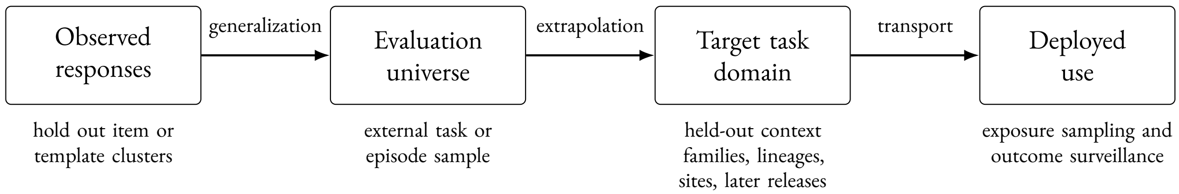

Figure 1: Where a score has to travel, and what answers each step. Each arrow is a separate claim with its own evidence, and warrant for one doesn't carry to the next. The lower row names the design that addresses the step above it; Table 1 gives the full set of relations.

Underrepresentation of the target is one inferential gap. Aggregation creates another even within a fixed evaluation universe. Zhang et al. (2026, pp. 2, 4–6) provide a controlled example: they add task-irrelevant context to benchmark items and find that aggregate accuracy often moves little even when item-level response probabilities move in opposite directions. After subtracting the change expected from repeated baseline sampling alone, their estimates reach 13.6% for mean absolute item change and 53.2% for worst-decile degradation. The affected examples are also largely model-specific: on MMLU-Pro the mean cross-model correlation of per-item change is .00, and the mean Jaccard overlap of the most-affected decile is .09.

A stable mean is compatible with unchanged items, balanced improvements and degradations, deterioration concentrated in a tail, or a persistent set of serious failures. Those states may support different predictions and decisions. The declared target identifies distinctions that may need to be preserved; matched evidence determines which earn a place in a target-specific report. Figure 2 sets the four side by side.

Interpreting a finite evaluation as evidence of general capability widens the gap between the standardized evaluation universe and the intended target. Raji et al. (2021) argue that broad benchmark claims routinely outrun the contextual tasks used to construct them. Hendrycks et al. (2025) propose a revisable framework for evaluating AGI across ten cognitive domains, guided by Cattell–Horn–Carroll (CHC) theory, a hierarchical account of human cognitive abilities inferred from test-score correlations. They warn that a total can conceal bottleneck failures and recommend reporting the domain profile, the vector of domain scores, alongside it.

The profile preserves distinctions that a total erases, but it still compresses item changes and tails, and its domain structure may not transfer across systems (Section 5.3). A score is an indicator, not the capability or attribute it's meant to represent; validity theory calls the represented capability or attribute the CONSTRUCT (Messick, 1995b, p. 742). Neither total nor profile establishes which inferences the ten-domain framework supports beyond its construction; a broader construct interpretation requires an additional argument (Kane, 2013, p. 24).

Before an aggregate is interpreted as a measure or used in a decision, three separate questions arise: whether its components share a meaningful scale (commensurability), whether the tradeoffs it permits are acceptable (compensability), and whether the resulting score supports the declared target inference (projectibility).

Claims labelled AGI are typically open-ended: no enumerated test set exhausts their intended range, and coverage alone can't validate unrestricted generality. The point isn't that a finite index is incoherent. A battery may stipulatively define an AGI index and report it accurately; the open-ended

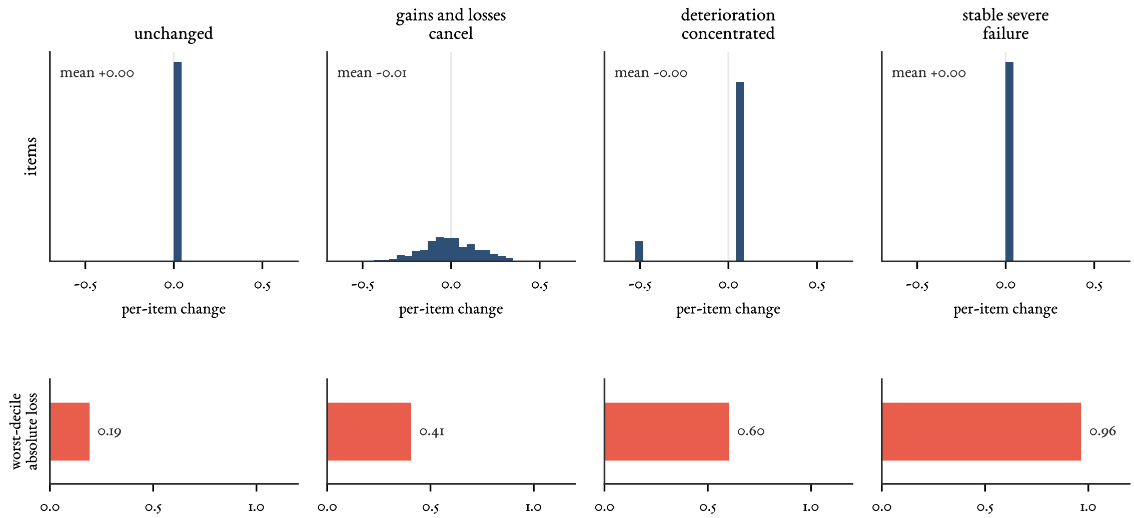

Figure 2: Four item-level states behind one stable mean, constructed for illustration rather than measured. The top row gives the distribution of per-item change; all four have the same near-zero mean. The bottom row gives the resulting worst-decile absolute loss. The first and fourth columns are indistinguishable in the top row and differ by .19 against .96 in the bottom one, which is why a change statistic can't stand in for an absolute risk measure.

claim begins when that index is interpreted as evidence of general capability beyond its construction domain. The developer's claim about the firm's assistant is of that kind, and no evidence the firm can gather will settle it. A study can instead identify a bounded source–target link, preserve distinctions potentially relevant to it, and combine empirical evidence with substantive or formal rationale where appropriate. The firm's route runs through such a link: a declared, testable claim about its own matters, which leaves the general capability question open. Aggregates can describe construction samples; broader uses require matching warrants.

Two adjacent proposals share the diagnosis and divide the labour differently. Liu et al. (2024) adapt evidence-centred design from educational assessment, formalizing benchmark construction into five modules and requiring designers to describe, justify, and support each choice, with case studies of BoolQ, SuperGLUE, and HELM. Their subject is the evidence a benchmark collects about the capabilities it declares; movement beyond the benchmark isn't their topic. Binette and Reiter (2024) adapt the estimands framework of the clinical-trials guidelines, characterizing an estimand by a metric, a scope or population, a data-acquisition strategy, and an aggregation rule, after which an estimator and its uncertainty are chosen. They do press how performance generalizes beyond a benchmark, showing how cross-validation and benchmark composition can mislead. What differs here is the organizing unit. Theirs is an estimand specified by metric, population, data acquisition, and aggregation. This paper's is an inferential chain whose links may call for different estimands, different holdout units, and different parties to supply the evidence.

Formal transportability asks a sharper version of the question for causal effects. Pearl and Bareinboim (2014) represent domain differences in selection diagrams and give conditions under which a causal relation in the target is uniquely computable from the source's observational and interventional distributions together with the target's observational distribution (their Definition 5). Transport is used more broadly below, covering non-causal movement across contexts, systems, sites, and time, and it doesn't inherit those identification results.

What this paper adds to those accounts is a combination: projectibility assessed for each source–target link rather than for an evaluation as a whole; each link tied to the facet it crosses and to evidence matched to that facet; behavioural-change quantities separated from target-conditioned loss; evidential responsibility divided between developer and deployer; and composition across links permitted only through an explicit interface argument.

This paper makes three linked contributions. First, it turns claims beyond the observed evaluation into explicit, separately testable source–target links and assigns responsibility for their evidence. Second, it shows why several kinds of behavioural change and target-conditioned risk can't generally substitute for one another. Third, it specifies a developer–deployer handoff for completing and maintaining warrants across the boundary between a reference evaluation and a local use. The reanalysis and simulations test the second contribution's statistical machinery; they don't validate an external deployment projection.

2 Projectibility within a Validity Argument

For AI evaluation, validity theory supplies the architecture for this proposal. Messick uses score broadly, for any qualitative or quantitative summary of observed regularities in people, groups, environments, objects, or institutions (Messick, 1995b, p. 741). Kane's argument-based approach represents a proposed score interpretation and use as an explicit chain of inferences and assumptions, then asks whether that chain is coherent, complete, and adequately supported. The more ambitious the interpretation or use, the more backing its chain requires (Kane, 2013, pp. 1–3). The same architecture applies to a base model, an agent, or a deployed model–tool–workflow composite, provided the bearer of the claim is explicit.

Within this architecture, projectibility is assessed for each source–target link. It's graded and claim-relative: the evidential support that specified observations and revisable background assumptions give a declared source–target link, relative to a proposed interpretation or use. Grading doesn't imply a single numerical scale; it blocks the reading that a link is projectible or not in isolation from the amount, quality, and defeat conditions of its evidence. It isn't a separate aspect of validity or a name for the warrant of an entire decision. A projectibility audit identifies each such link, states the facet over which the claim is meant to travel, matches evidence to that facet, assigns responsibility for supplying the evidence, and blocks successful bounded links from automatically composing into a broader claim.

For confirmatory use, the audit begins with a prospective declaration. A claim formulated after inspection of source results can still be tested on genuinely fresh evidence, but the source analysis can't confirm that claim. The declared target also has to differ from the source condition. A claim that projects a benchmark result to tasks individuated by that benchmark's own formats returns its admission criterion, so it earns no extrapolative credit beyond the declared evaluation universe. Generalization within that universe remains a genuine inference, and the first row of Table 1 treats it as one.

Prospective declaration narrows discretion without eliminating it. Multiple comparisons can distort inference even where there was no fishing and the hypothesis was fixed in advance, because the analysis stays contingent on the data in ways a declaration doesn't fully determine (Gelman & Loken, 2013). A declaration makes the remaining discretion inspectable; it doesn't remove it.

The audit also keeps four questions apart: what projective claim is being made; what evidence supports it; what regularity or causal dependence makes the projection reliable; and what, if anything, explains that dependence. Evidence can support a prediction without identifying its cause, and a proposed explanation can be plausible without yet licensing projection to a new population.

A decision adds another layer. Projectible factual premises have to be combined with feasible alternatives, constraints, institutional authority, values, and an action rule. Values also enter construct labels, content definitions, and score interpretations rather than appearing only at the final decision stage (Messick, 1995b, pp. 747–749).

Projectibility covers only the beyond-observation links in the larger validity argument. Messick also asks whether content represents the intended domain, observed responses engage the relevant processes, the scoring structure fits the claimed construct, relations with other measures and criteria support the interpretation, and consequences bear on the proposed use (Messick, 1995b, pp. 741–747). No holdout design answers all of those questions. Nor are the projective links interchangeable: generalization within a stated evaluation universe, extrapolation to a broader task domain, transport across contexts, systems, or populations, and temporal prediction require different evidence.

System transport requires one further term. Here, LINEAGE denotes an operational dependency group defined by declared provenance, training, distillation, shared components, or data dependence rather than by a family label alone. Outputs from closely related versions don't supply the same evidence as independent lineages. Lineage is often only partly observable: training-data overlap, distillation, and shared components are frequently proprietary, so an assumed independence between systems may itself be an undeclared assumption.

Table 1 distinguishes eight inferential relations and then separates them from decision use. Each row poses a different evidential question within one interpretation-and-use argument.

Table 1: Inferential relations, decision use, and especially relevant evidence.

Claim or use	Declared range	Especially relevant evidence
Observed responses to evaluation universe (generalization)	Declared item types, templates, and repeated-response conditions	Representative sampling, holding out whole item or template clusters, or a design that estimates variation across the declared sources
Evaluation universe to target task domain (extrapolation)	Intensionally specified tasks and operating conditions	External task or episode sampling plus content and process rationale
One context to another	Realizations or perturbation families	External realizations or families and explicit assumptions about what remains comparable
One system population to another	Models, lineages, or configurations	Independent systems or lineages plus prespecified comparability assumptions or models for transfer
One operator, workflow, or site to another	Roles, organizations, workflows, or sites	External target sampling, checks for target features that alter results, and justified sharing of information across sites or groups
Present to future behaviour	Sessions, releases, or deployment periods	Chronological observations at the stated horizon plus temporal rationale
Behaviour to causal explanation	Processes, mechanisms, and rival routes	Process evidence, interventions where the claim and access permit them, and rival tests
Behaviour to consequence distribution	Affected parties, exposure paths, and settings	Exposure sampling, outcome surveillance or impact study, and analysis by affected group
Evidence and values to decision	Actions and consequences	Calibrated factual premises, feasible alternatives, declared values and constraints, distributional analysis, and monitoring

The running law-firm example shows how several of these relations combine in one bounded decision. Two source–target links run off the same published battery. The developer's link, from multidomain scores to broad capability, spans no declared range and names no defeating observation, so it isn't the kind of claim this audit can assess. The firm declares the other: that the next version of the complete model–retrieval–tool workflow will keep its worst-decile case risk at or below .20 when local matters contain an unseen family of answer-irrelevant context. Case risk is expected loss on a prospectively specified 0–1 expert rubric for matter-level research quality.

The joint target distribution weights matter episodes by projected use across declared matter types and jurisdictions, with lawyers as operators and clients, courts, and third parties recorded as affected populations. Source and target are partitioned differently: the battery reports cognitive domains, the firm's rule is conditioned on practice areas, and neither partition refines the other. Domain-level source results don't map onto practice-area tails without an argument. The .20 rule applies to the maximum of practice-area-conditional tails, together with separate red-line constraints, rather than to a pooled loss across affected parties. The horizon is the next release cycle.

Earlier versions and familiar context families form the source evidence. The developer supplies model provenance, reference prompts, scorers, and perturbation results; the firm supplies the local task and exposure distribution, rubric, alternatives, and constraints. A later version tested on an untouched context family and held-out local matters supplies the final test of the rubric-scored claim. The firm reports the estimated maximum practice-area tail with uncertainty required to account for the selection of that maximum, for example through a simultaneous bootstrap over the practice-area tails (Section 3.7). The .20 figure is a declared constraint, and the decision needs the posterior probability that the later release's tail exceeds it, together with the losses of the actions available: release, restricted use, mandatory review, or no deployment. That's a statement about one release's fixed risk parameter. A predictive probability of exceedance in some future release period is a different quantity, and Section 5.1 sets out what it further requires. An interval-crossing rule is one possible action rule, and every rule mapping continuous evidence to a discrete action has a boundary somewhere; expected-loss minimization also switches actions discontinuously. The objection is narrower: such a rule imports an error-control convention, and with it an implicit loss function, instead of using the losses, constraints, and alternatives the declaration has already specified. Report the estimate and its uncertainty continuously, and justify whatever action rule is adopted against those declared quantities.

The rubric is a surrogate: its relation to realized consequences for clients, courts, and third parties is a separate projection that held-out task scoring doesn't establish and that requires exposure and outcome surveillance in use. A random item split within familiar contexts tests neither the version nor context transport at issue.

The example instantiates a general requirement. For any inferential row in Table 1, and for any decision that uses it, a confirmatory evaluator should register a PROJECTION DECLARATION before calculation. An exploratory analysis should complete the same fields and reserve new evidence for confirmation:

1. Claim and source–target relation: the outcome, interpretation, or explanation to be inferred, the source evaluation universe, the bounded target range, and the minimum reach that would serve the intended use
2. System population: the model families, versions, configurations, and technical settings in scope

3. Operator and affected populations: the roles, teams, and sites participating in performance; the people or groups exposed to its consequences; and any relevant exposure strata
4. Task population: the intensional task specification or admissibility rule and its sampling frame
5. Bearer and unit: the base model, agent, or deployed model–tool–retrieval composite and the observational unit
6. Intervention range: the perturbation families and realizations over which the claim is meant to hold, which have to cover the variations the intended use makes foreseeable rather than only those that make success likely
7. Time horizon: immediate, delayed, or later deployment behaviour
8. Risk measure: the item, episode, or realization indexing loss; its consequence bearer or bearers and exposure measure; the joint target-population probability distribution; whether the tail is taken over conditional case risks or realized losses; the tail fraction and minimum tail count; and whether the rule applies jointly, by stratum, or to the maximum conditional tail
9. Decision criterion: the objectives, consequences, feasible actions and alternatives, preferences or loss, applicable constraints, and action threshold
10. Evidential standard, failure conditions, and responsibility: the evidence and tolerance that could support the claim, what would defeat or restrict it, what would leave it unresolved, any fallback scope to be tested if it fails, who supplies the source evidence, and who can perform target-specific testing

When the source evaluator (typically the developer) and the target decision-maker (the deployer) are different organizations, the declaration assigns responsibility for each part of the source–target warrant without treating either party's evidence as sufficient for the whole.

Declaring a claim prospectively doesn't stop it from being rescued after failure. A defeated claim may be retested at a fallback scope only if that scope was fixed in advance, the contraction is reported as a loss of generality, and the narrower claim is assessed on evidence that didn't inform the revision. A scope drawn around the cases that happened to survive is a new claim needing a new test. When the fallback falls below the declared minimum reach, the projection is retired for that use rather than narrowed again.

The projection declaration states what has to be warranted. A separate sampling audit records what generated the observations: model, version, and lineage; item and template cluster; domain; perturbation family and realization; operator role; team or site; affected-population and exposure stratum; response replicate; scorer; and time. Measurement theory calls each potentially varying part of this design a FACET.

Facets are CROSSED when their levels combine and NESTED when one varies only within another. Fixed levels are those studied; sampled levels stand for a population. GENERALIZABILITY THEORY studies how facets affect score meaning and precision (Brennan, 2001).

Sampling coverage isn't the only issue. Representative task sampling and representation of the intended processes are different obligations: one concerns coverage of the performance domain; the other concerns whether the observations engage the processes and structures named in the interpretation (Messick, 1995b, p. 745). More sampling from a poorly represented process doesn't repair the process claim, and a well-supported process account doesn't establish population coverage.

A facet interaction may be error for one projection and signal for another. Variation that depends jointly on system, item, and context is error for a context-invariant ranking but signal for predicting context-specific risk. Variation across system–operator combinations may be nuisance for a base-model claim and essential signal for a workflow claim. Messick's analogous example is reading demand: irrelevant to one subject-matter interpretation but relevant to a criterion that itself requires reading (Messick, 1995b, p. 743). The model-specific item changes reported by Zhang et al. (2026, pp. 2, 4–6) show why the classification has to follow the target.

3 Distinctions Lost under Aggregation

Aggregation can conceal different things at different levels: shifts in a domain profile, offsetting item changes, concentrated deterioration, serious failures that persist without change, and, once losses are in view, which parties bear them. This section studies those possibilities in PAIRED-CONDITION EVALUATION, in which the same items are scored under a baseline and an altered condition. The sequence moves from profile shape and level to itemwise change, change tails, and absolute loss. Each addition answers a question that the preceding quantity can't.

The quantities form a diagnostic menu, not a universal dashboard. A target-specific study should preregister the subset that bears on its claim and specify which sense of *stable* is intended. Publishing the full module can clarify stability in several senses, but it doesn't make the claim unrestricted. Retaining the paired source data keeps omitted distinctions auditable.

3.1 Data Structure and Three Targets of Estimation

Fix a system, time, scorer protocol, and perturbation realization. Let domains be indexed by $g = 1, \ldots, G$, items within domain g by $i = 1, \ldots, N_g$, and conditions by c, with baseline 0 and perturbation p. Let $s_{igc} \in [0, 1]$ be the item score. In a sequential-agent evaluation, an item may be a complete task episode or trajectory, provided its score and target loss are defined at that level. Define

$$\delta_{igp} = s_{igp} - s_{ig0}, \qquad a_{gc} = \frac{1}{N_g} \sum_{i=1}^{N_g} s_{igc},$$

and let $\mathrm{a}_c = (a_{1c}, \ldots, a_{Gc})$.

An ESTIMAND is the quantity an analysis aims to estimate. The same symbols below support three readings. First, with deterministic decoding on a fixed paired set, the quantities are exact descriptions of that set. Second, under a declared stochastic generation and scoring protocol, s_{igc} is an expected score estimated from repeated responses and, when relevant, scorers.

Third, inference to an item, context, system, or time population requires sampling or holdout from that declared universe. Repetition within an item doesn't create a representative item sample, and counting response rows as independent observations would generally be wrong. The independent unit follows the projection declaration.

For a population estimand, unequal item inclusion probabilities require design weights, and a q-tail should represent q of the target-population mass. The unweighted formulas below use the empirical item measure, so they describe a fixed equally weighted set or an equal-probability sample.

Table 2 gives the paired-condition module used here. Reports using it should retain the raw domain profiles and item-change distributions alongside the summaries.

Table 2: Paired-condition behavioural descriptions and target-conditioned risk.

Output	Granularity	Question answered
r_p (profile correlation)	Domain profile	How strongly were the centred, scaled domain profiles linearly associated?
L_{gp}, L_p (signed level change)	Domain and overall level	Did average performance rise or fall?
INS_{gp} (item instability)	Items within one domain–condition pairing	How much two-sided paired item change occurred?
F^+_{gp}, F^-_{gp} (directional components)	Items within one domain–condition pairing	How much of that change was improvement and how much degradation?
$\mathrm{WTD}_{q,gp}$ (worst-tail degradation)	Change tail	How large was mean degradation in the worst-changing q-fraction?
Absolute-loss tails	Cases and realized draws	How large were the case-risk and realized-loss tails?

3.2 Profile Correlation and Signed Level

For nonconstant profiles, report Pearson correlation directly:

$$r_p = \mathrm{corr}_g(\mathrm{a}_0, \mathrm{a}_p) \in [-1, 1].$$

A high r_p establishes only strong linear association between the centred, scaled profiles. A common shift and positive rescaling of the profile can yield $r_p = 1$, even when level or spread changes.

With ten fixed domains, r_p is a descriptive summary of those ten estimated means. Resampling items within domains propagates uncertainty in the means, but not uncertainty in the choice of domains. Correlation is undefined when either profile has zero variance and equals either -1 or 1 when there are only two domains and both profiles are nonconstant. Reports should publish the profiles and their spreads; a preregistered centred distance or rank correlation may supplement r_p when it better matches the target.

Because correlation discards level, report signed domain and overall changes separately:

$$L_{gp} = a_{gp} - a_{g0}, \qquad L_p = \frac{1}{G}\sum_{g=1}^{G} L_{gp}.$$

They lie in $[-1, 1]$ for unit-interval scores. The equal-domain mean describes an equally weighted domain set; alternative domain or deployment weights encode different scales and target mixtures and have to be labelled and justified.

3.3 Item Instability and Directional Components

Signed change can cancel within a domain. Mean absolute paired change is

$$\mathrm{INS}_{gp} = \frac{1}{N_g}\sum_{i=1}^{N_g} |\delta_{igp}| \in [0, 1]. \tag{1}$$

Its directional components are

$$F_{gp}^{+} = \frac{1}{N_g} \sum_i (\delta_{igp})_+, \qquad F_{gp}^{-} = \frac{1}{N_g} \sum_i (-\delta_{igp})_+.$$

Here F^+ is mean positive change and F^- is mean negative change, recorded as a nonnegative magnitude. For deterministic binary scores, these are the incorrect-to-correct and correct-to-incorrect transition proportions. The identities

$$L_{gp} = F_{gp}^{+} - F_{gp}^{-}, \qquad \mathrm{INS}_{gp} = F_{gp}^{+} + F_{gp}^{-}$$

make cancellation explicit and imply $|L_{gp}| \leq \mathrm{INS}_{gp}$.

3.4 Worst-Tail Degradation

INS averages change magnitude over all items; it doesn't show whether deterioration is concentrated. To inspect the worst-changing tail, choose $q \in (0, 1]$ and a minimum tail count before inspecting results. Let $m_g = \lceil qN_g \rceil$, and order paired changes from smallest to largest. Define

$$\mathrm{WTD}_{q,gp} = -\frac{1}{m_g} \sum_{\ell=1}^{m_g} \delta_{(\ell)gp} \in [-1, 1]. \tag{2}$$

This expression averages degradation among the worst-changing q-fraction. A positive WTD indicates mean deterioration among the selected items; a negative value means that even the selected lower tail improved on average.

In risk analysis, EXPECTED SHORTFALL is the average loss in a prespecified worst fraction of cases. WTD is its finite-sample analogue applied to the loss $-\delta$ (Acerbi & Tasche, 2002; Rockafellar & Uryasev, 2002). Because m_g rounds qN_g upward, the empirical tail mass is m_g/N_g; the selected set is exactly a q-tail only when those quantities coincide. Since a selected lower-tail mean can't exceed the full mean, $\mathrm{WTD}_{q,gp} \geq -L_{gp}$, with equality at $q = 1$. WTD measures signed worst-tail change; consequences enter through the target-specific loss defined below.

3.5 Absolute Worst-Tail Loss

A change statistic misses a stable serious failure. If the same safety-critical items are failed at baseline and under perturbation, their δ_{igp} values are zero, so both INS and WTD can be zero. The law firm's release decision still needs an absolute loss constraint, whose decision-facing form appears in Section 3.6.

For target T, let R collect response, scorer, and outcome randomness within an item–condition pairing, let $\ell_{igcR}^{(T)} \in [0, 1]$ be a realized item or episode loss, and let $\mu_{igc}^{(T)} = \mathrm{E}_R(\ell_{igcR}^{(T)} \mid i, g, c)$ be its conditional expected loss.

Both realized loss $\ell_{igcR}^{(T)}$ and conditional expected loss $\mu_{igc}^{(T)}$ may be prospectively scored surrogates rather than observed downstream consequences. That status and any proposed surrogate–consequence link have to be stated. Because WTL averages loss, differences on the chosen scale have to carry the intended meaning (Keeney & Raiffa, 1993). Ordinal hazard classes should remain separate strata or define constraints rather than be averaged as if their labels were cardinal.

Order condition-specific case risks from largest to smallest and define

$$\mathrm{WTL}^{\text{case},(T)}_{q,gc} = \frac{1}{m_g} \sum_{\ell=1}^{m_g} \mu^{(T)}_{[\ell]gc} \in [0,1]. \tag{3}$$

Equation 3 identifies cases with high conditional expected loss. An empirical realized-loss tail instead orders draws of $\ell^{(T)}_{igcR}$ from a specified mixture over item–condition pairings and residual realizations. The two coincide only under restrictive conditions, such as no residual loss variation within pairings. Report the baseline and perturbed levels of whichever tail the declaration selects as primary results and the difference between those conditions as a secondary result.

A binary $1-$correctness loss is inadequate when consequences differ across items. If loss isn't normalized, report it in its natural units. Equation 3 is a domain-conditional diagnostic; a decision-facing tail uses the declared joint target distribution or an explicitly chosen set of conditional constraints.

3.6 From Fixed Batteries to Target-Population Risk

The finite-set formulas weight observed benchmark items equally; they don't by themselves define an action-level risk distribution. A decision-facing tail instead needs a probability distribution over the cases the decision concerns. The evaluator has to specify whose cases, which conditions, and what losses define the population being ordered.

For a deployment-facing target, let $J \sim P_T$ index a joint draw over every varying facet in scope, including system, task or episode, domain, perturbation family and realization, operator, affected-population stratum, and time. Let R collect response, scorer, and downstream-outcome randomness conditional on J, let S_{JcR} be the realized score under condition c, and let $\ell^{(T)}_{JR} \in [0,1]$ be realized target loss. Define $\delta_J = \mathrm{E}_R(S_{JpR} - S_{J0R} \mid J)$, $D_J = -\delta_J$, and $\mu^{(T)}_J = \mathrm{E}_R(\ell^{(T)}_{JR} \mid J)$.

With $Q_{V,T}$ the quantile function of V, which gives the cutoff at cumulative probability u under the indicated target distribution, define

$$\mathrm{ES}^{+}_{q,T}(V) = \frac{1}{q} \int_{1-q}^{1} Q_{V,T}(u)\, \mathrm{d}u.$$

The deployment-facing change tail and absolute-loss tails are

$$\mathrm{WTD}_{q,T} = \mathrm{ES}^{+}_{q,T}(D_J), \qquad \mathrm{WTL}^{\text{case}}_{q,T} = \mathrm{ES}^{+}_{q,T}(\mu^{(T)}_J), \qquad \mathrm{WTL}^{\text{real}}_{q,T} = \mathrm{ES}^{+}_{q,T}(\ell^{(T)}_{JR}).$$

The expected-shortfall functional orders probability mass rather than merely similar-looking benchmark items. When the distribution is discrete and the cutoff falls within a probability mass, the tail calculation includes only enough of that mass to reach the chosen fraction.

The realized-loss measure integrates both $P_T(J)$ and the conditional distribution of R, so it generally differs from the tail of conditional case risks. A decision has to name which tail it constrains and whether the rule applies jointly, by stratum, to the maximum conditional tail, or through separate limits that gains elsewhere can't offset. If repeated exposure matters, let J index a person–horizon or task trajectory and report cumulative exposure separately.

3.7 Estimation, Selection, and Multiplicity

Those formulas can describe fixed sets exactly, but estimating latent expected scores or population tails from stochastic generation or scoring introduces two nonlinear problems: absolute values create a positive response-sampling floor, while selecting the worst observed tail selects noise. Selection on noise exaggerates magnitude, so a selected tail overstates its latent value: a Type M error in the sense of Gelman and Carlin (2014). The positive-part directional components F^+ and F^- inherit the same floor. The baseline-only PSEUDO-NULL procedure used by Zhang et al. (2026) is a resampling scheme that simulates no condition difference using baseline responses. It estimates the nonlinear statistic expected under that simulation.

Subtracting the pseudo-null expectation from the raw statistic yields a useful null-referenced diagnostic, not a general bias correction: under a real effect, response variances and the selection floor can change nonadditively. If reported, the pseudo-null contrast should reproduce the real response counts and remain untruncated, with uncertainty.

When latent item changes are the target, the cleanest treatment models items hierarchically and lets partial pooling shrink noisy extremes toward the domain mean, so the tail becomes a summary of the fitted item effects rather than a set selected on noise. That also dissolves the estimand ambiguity described next. Where a model isn't wanted, tail selection and estimation should at least use disjoint response splits. CROSS-FITTING repeats split-sample selection and estimation with the roles of the splits reversed, avoiding reuse of the same response noise. It targets the latent effect of the selected set, which may differ from the latent population tail. Deterministic fixed-set descriptions have no response-sampling contribution to remove, although uncertainty about new items or contexts remains.

Response-noise corrections don't address projection across facets. Resampling and holdout units should follow the projection declaration, preserving paired items, shared baselines, perturbation realizations, scorers, and model lineage. An item bootstrap conditional on fixed domains and perturbations treats items as nested within those fixed facets and crossed with conditions, scorers, and replicates, so it estimates item-sampling uncertainty within them. Claims about new domains, perturbation families, or systems require those levels to be sampled or held out in turn.

For case-risk WTL, uncertainty can also enter through conditional loss estimates, target probabilities, severity elicitation, exposure weights, and sparse group samples. Realized-loss WTL also requires the response, scorer, and outcome variation represented by R. Reports should propagate those sources, distinguish intervals for individual comparisons from intervals or error rates that account for all reported comparisons together, and state how the analysis accounts for examining many tails or subgroups.

Even perfectly estimated summaries don't repair a bad benchmark. Disaggregation can't resolve ambiguous items or an ill-designed perturbation. Annotation quality, statistical power, construct coverage, and the property that the stress test is supposed to leave unchanged have to be justified independently (Bowman & Dahl, 2021).

3.8 Behavioural Description and Target-Conditioned Risk

Profile shape, signed level, item instability, directional change, and WTD are behavioural descriptions. By contrast, the WTLs summarize tails of conditional case risk and realized loss under a declared loss scale, exposure distribution, and affected population. These quantities answer distinct questions and needn't be manifestations of one latent variable.

A high r_p with negative L_p means that the centred, scaled profiles remained linearly associated while level fell. Near-zero L_{gp} with high INS means that changes cancelled. High WTD localizes degradation in a change tail, while a high absolute-loss tail can reveal stable risk even when every change statistic is zero.

When this paired-condition module is used, publish the behavioural report separately from any target-conditioned risk report. A scalar may summarize the construction sample when its operation, scale, and weights are explicit. Any warrant established for a score interpretation, predictive model, or decision rule is limited to the target and population represented by its evidence.

4 Empirical Demonstration and Estimator Checks

This section performs two checks. First, a reanalysis of released trial-level outputs from Zhang et al. (2026) verifies the paired-condition calculations on fixed data. Second, simulations with known latent values compare raw, null-referenced, and split-sample tail estimators, distinguish change tails from case-risk tails, and probe profile correlation when between-domain spread is low. The empirical companion at https://github.com/BrettRey/agi-evaluation-hpc contains the full analyses, including data-generating processes, replication counts, and estimator implementations; Figure 4 summarizes them.

4.1 Released-Trial Reanalysis

The released design crosses four benchmarks with eight models, yielding 32 comparisons with 20 baseline and 20 context trials per item (Zhang et al., 2026, pp. 3–4). The reanalysis uses the released responses rather than regenerating outputs from proprietary systems. The companion pins the preprint version, repository commit, data hashes, seeds, and analysis settings. It's verification on already published results, not a confirmatory test in the sense of Section 2: it was specified after those results were public and claims no prospective standing. Its quantities are estimated in the stochastic-response mode of Section 3.1, with models and benchmarks as fixed levels.

Recomputed values agree with the rounded table in the preprint to within .20 percentage points across baseline accuracy, signed change, INS, and WTD. This agreement verifies reproduction of the published pipeline.

For MMLU-Pro–gpt-5.4, baseline accuracy is .8119 and context accuracy .7904, a signed change of -.0215 with a 95% item-bootstrap interval of $[-.0360, -.0070]$. Raw worst-decile WTD is .3680, interval $[.2870, .4510]$; subtracting the baseline pseudo-null expectation of .1049 gives a null-referenced value of .2631, and the response-half split estimate is .2911. Thus a 2.15-percentage-point average decline coexists with much larger change in the selected tail. These intervals resample items within fixed domains and perturbations, so they carry item-sampling uncertainty alone. Benchmark items weren't probability-sampled from any stated universe, so the intervals are model-based, resting on an exchangeability assumption about items of the declared type, rather than design-based. The tail interval needs a further warning: it brackets the raw statistic at the released trial count, which is itself inflated by selection on response noise, so it isn't an interval for the latent worst-decile effect. Section 4.2 shows that it covers that latent quantity essentially never. The null-referenced and split estimates, not the interval, are what address the inflation.

The signed change also conceals two-sided movement. Its directional components are $F^+ = .0229$ and $F^- = .0444$, which sum to the item instability of .0673 and differ by the signed change. In 22 of the 32 released comparisons both components exceed twice the absolute signed change, so most cells show substantial movement in both directions behind a small net. As with every count and average over these cells, that tally describes one fixed crossed dataset with shared items and lineages, not 32 replications.

The same comparisons also separate a change tail from an absolute one. Cross-fitted worst-decile case risk runs from .924 to .996 across the 32 cells while the split change tail runs from .039 to .617. Under a binary $1 -$ correctness loss at these accuracies a high case-risk tail is close to arithmetic, so this isn't a finding about these particular models. It shows that the two quantities come apart outside simulation, and that a change statistic can't be read as an absolute one. Figure 3 shows both contrasts.

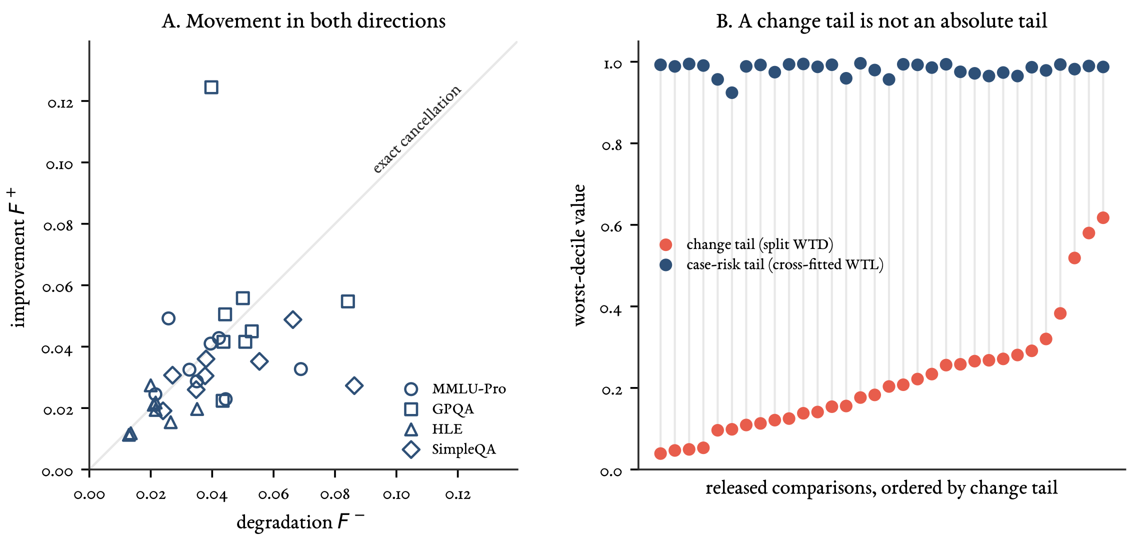

Figure 3: Two distinctions the released aggregates conceal. (A) Directional components for the 32 comparisons. The diagonal is exact cancellation, where improvement and degradation are equal and the signed change is zero; distance from the origin is how far items moved, and distance from the diagonal is how much of that movement survived aggregation. (B) The change tail and the absolute case-risk tail for the same comparisons, ordered by the change tail. Both estimators use disjoint response information for selection and estimation, the split estimator for the change tail and cross-fitting for the case-risk tail, so the comparison isn't confounded by reuse of the same response noise. Neither thereby recovers an ideal latent population tail. Under a binary loss at these accuracies the case-risk tail is necessarily high; the point is that neither quantity determines the other or can substitute for it, not that they're statistically unrelated in every population.

The null-referenced and split-tail procedures estimate different quantities. Split WTD exceeds the null-referenced value in 31 of the 32 released comparisons, by .0331 on average. Those comparisons share items within a benchmark and lineage across models, so the tally describes the fixed released set rather than 31 independent replications. Selecting and estimating the tail on the same responses inflates it: the raw estimate exceeds the split estimate by .0906 on average, ranging from .0330 to .1696. The null-referenced procedure subtracts a baseline-null expectation from a statistic selected and measured on all trials; it needn't remove bias under a nonnull effect. The split procedure selects items on one response half and measures their changes on the other, estimating the latent effect of a noisy-selected set rather than the ideal tail selected using unobserved latent item effects.

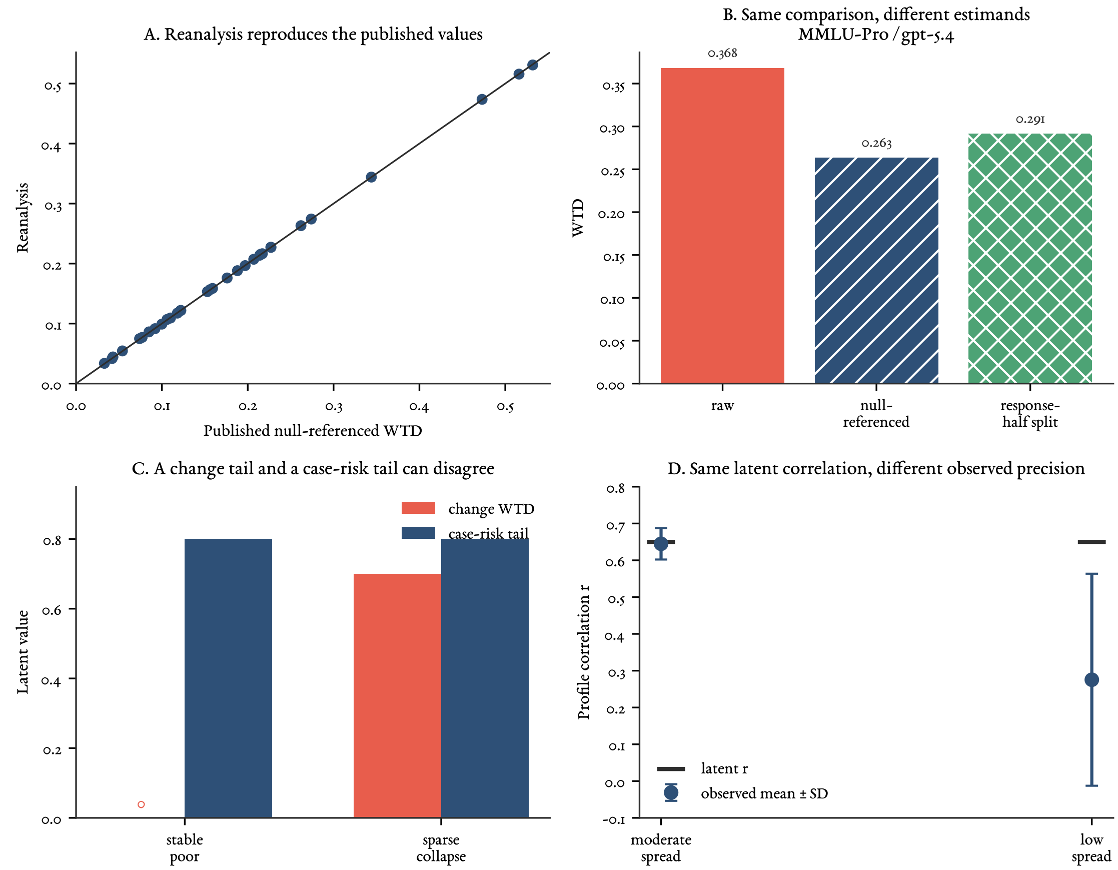

Figure 4: Released-output reproduction and simulation checks. (A) Recomputed null-referenced WTD for all 32 comparisons against the rounded values reported by Zhang et al. (2026, p. 4). (B) Raw, null-referenced, and response-half split WTD for MMLU-Pro–gpt-5.4; the latter two target different estimands. (C) A simulation with stable, persistently poor performance on the same items has zero latent WTD but .80 worst-decile latent itemwise expected loss (case-risk WTL). (D) With ten domains and latent $r = .65$, correlation estimation remains accurate at moderate profile spread but attenuates under low spread. Full settings and numerical outputs are in the empirical companion.

4.2 Known-Truth Checks

Because simulations fix latent item probabilities, they make each estimator's target observable. Under a null effect, mean null-referenced INS, null-referenced WTD, and split WTD are .0033, .0039, and -.0018 against a truth of zero. Depending on the estimator, 31–56% of the estimates are negative. Keeping those values untruncated is necessary for a zero-centred diagnostic.

In the sparse-collapse scenario, 10% of items deteriorate by 55 percentage points while the remainder are unchanged. Null-referenced WTD averages .3445 against the ideal worst-decile effect of .5500 selected from the latent truth. Split WTD averages .4514 against a true mean of .4550 for the sets selected on noisy response halves, a bias of -.0036. Cross-fitting nearly removes reuse bias for its selected-set estimand, which differs from the ideal latent-tail estimand. Reports should state which tail they estimate.

In the stable-poor scenario, the same low-performing items have unchanged response probabilities across conditions, so latent WTD is zero. Their worst-decile latent itemwise expected loss is .80, the case-risk WTL. The tail of realized binary errors is a different estimand. Cross-fitted WTD and case-risk WTL are .0013 and .7911. A change tail and an absolute case-risk tail can disagree even without sampling error; the disagreement reflects their definitions.

The item bootstrap needs its own known-truth check, because a tail interval invites a reading it can't support. With 500 items, 20 trials per condition, and heterogeneous latent effects, 95% item-bootstrap intervals cover the population signed change in .94 of replicates and population INS in .93. For the raw worst-decile statistic at that trial count they cover in .92, the mild undercoverage typical of a tail functional. For the latent worst-decile effect they cover in .00. Selection on response noise raises the population raw statistic from .160 to .295 while the interval is .056 wide, so the two never meet. An item bootstrap quantifies item sampling; it doesn't correct selection, and the null-referenced and split estimates are what address that.

The final simulation returns to the profile-correlation caution in Section 3.2. With ten domains and latent $r = .65$, moderate between-domain spread yields a mean observed $r = .645$ and root-mean-square error .043. Holding the latent correlation fixed while reducing profile spread lowers the mean observed correlation to .275 and raises root-mean-square error to .472. Coverage of 95% bootstrap intervals that resample both domains and within-domain items falls from .988 to .579 even as mean interval width increases from .249 to 1.111.

When the perturbed latent profile is flat, its true correlation is undefined, but finite samples produce correlations with standard deviation .324. Reports should accompany r with the raw profiles and their spreads.

The reanalysis verifies the calculations on 32 fixed comparisons, and the simulations establish conditional estimator behaviour under their specified designs. Neither tests any inferential relation in Table 1; each of those requires a separate empirical design matched to the declared target, as Section 5 sets out. For the law firm, results of this kind show which distinctions a developer's report has to preserve; they say nothing about whether a published battery result holds on local matters.

5 Testing Declared Projections

Once source descriptions are fixed, each declared source–target link needs evidence matched to its own inferential relation. Evidence about test content, response processes, score structure, comparability across systems, prediction, causation, and decision consequences supports different premises in the validity argument.

Predictive projection provides the clearest illustration of facet-matched validation, so it comes first. The remaining subsections ask when an aggregate has a coherent interpretation, whether CHC-based interpretations transfer to AI systems, and what decision use adds.

5.1 Predictive Projection: Match the Statistical Design to the Claim

For a predictive projection, each row of the analysis should represent the independent unit fixed by the declared source–target link. To generalize to new items, hold out task-template clusters. To extrapolate to a broader task domain, draw an external task or episode sample rather than more items from the same universe. To transport across perturbations, hold out whole context families within systems. To transport across system populations, hold out lineages rather than outputs. To transport across workflows or sites, hold out whole sites or teams rather than operators within one site. To predict future behaviour, split chronologically. Response replicates estimate within-unit quantities; they don't multiply the number of independent systems, items, or contexts.

Let u index the independent unit appropriate to a declared projection, Y_u the external outcome, and X_u a preregistered set of baseline and granular predictors constructed without Y_u. Compare a simple model h_0, such as equal-domain baseline performance or a prior incident rate, with a richer model $h_1(\mathrm{X}_u)$. Under a target prediction loss $\mathcal{L}_T$, which scores forecasts of Y_u and is distinct from the item loss $\ell^{(T)}$ of Section 3.5, the relevant increment is

$$\Delta_T = \mathrm{E}_{\text{holdout}}[\mathcal{L}_T\{Y, h_0\} - \mathcal{L}_T\{Y, h_1(\mathrm{X})\}] \,.$$

A positive estimate with decision-relevant uncertainty supports added predictive value within that holdout population. It doesn't establish transport to another population.

Shared baseline observations make perturbation comparisons dependent; related versions share lineage; and granular predictors are estimated with error. When many comparisons predict outcomes for few independent systems, analysts may need to constrain the fitted model or reduce the number of predictors. Models for repeated measurements can address dependence, hierarchical models can share information across groups, and methods for predictors measured with error can represent uncertainty in those predictors. Reports should state the observational row, dependence model, predictor construction, and final holdout.

For the running release decision, model fitting and threshold setting use earlier versions and context families, followed by assessment of a later version under an untouched context family and on held-out local matters. Thresholds, tail fraction, feature selection, and model tuning should be fixed without examining the final test set. That final test can supply an interval for the maximum practice-area-conditional case-risk WTL of the later release; it can't by itself calibrate a forecasting model.

Forecasting future release-period WTL requires repeated independent release-period or system-level units. A model of the exceedance event, the maximum practice-area tail rising above .20, also requires calibration and scoring of that event probability. A model of case risk, a model of realized loss, and a model of a tail exceedance are distinct predictive objects.

Incremental predictive value, coverage or error for the tail measure, and performance of the resulting policy are three separate judgments. Evaluate the final holdout using the preregistered support, defeat, and inconclusive rules from the projection declaration; those rules should reference the estimated quantity and its declared tolerance rather than merely the position of an interval relative to a threshold. The evidential report shouldn't collapse continuous evidence into a crossing; an action rule may use one, provided its error costs and the available alternatives justify it. Too few independent lineages or severe events can leave the claim unresolved rather than show that transport failed. A defeated claim moves to its prespecified fallback scope if it has one; it isn't repaired by narrowing the target after seeing which cases survived.

5.2 Commensurability, Compensability, and Projectibility

Predictive usefulness doesn't by itself make a collection of predictors a coherent score. Aggregation adds questions about meaning and tradeoffs. Commensurability asks whether differences on component scales can be combined with the claimed meaning. Compensability asks whether strength in one component may offset weakness in another for the proposed use. PROJECTIBILITY asks whether the resulting summary supports the declared target inference. Messick (1995b, pp. 745–747) treats the choice between compensatory totals, profiles, and minimum requirements as part of asking whether a score's structure matches its intended interpretation; Carroll (1993, p. 632) notes that linear factor models assume compensation.

A monotone score such as

$$S_T = \sum_{g=1}^{G} w_g^{(T)} a_{g0}, \qquad w_g^{(T)} \geq 0, \quad \sum_g w_g^{(T)} = 1$$

retains an ordinary direction: more of each component can't lower the total. It also encodes substitution among domains over the stated score ranges. Mapping heterogeneous scores to $[0, 1]$ doesn't establish that equal differences have equal meaning or that those substitutions are acceptable. In a value model, the weights represent range-dependent tradeoffs rather than context-free domain importance (Keeney & Raiffa, 1993).

A predictive model may legitimately use negative conditional coefficients, interactions, nonlinearities, or redundant predictors. Those coefficients needn't express value tradeoffs. An interpretable aggregate, by contrast, shouldn't separately weight algebraically dependent quantities such as L, INS, F^+, and F^-, because the redundancy obscures the effective coefficients on positive and negative change. WTD is bounded by the same components and coincides with $-L$ at $q = 1$. Behavioural descriptions and target-conditioned risk shouldn't enter one interpretable total at all (Section 3.8). Equal-domain weighting remains a useful descriptive and predictive baseline because it's transparent and difficult to tune, but equal nominal weights needn't make equal statistical contributions when domain variances and covariances differ (Brennan, 2001, pp. 305–308).

Interpreting an aggregate as a measure requires evidence that its tasks cover the intended domain, responses engage the relevant processes, score structure matches the construct, and scores remain comparable across systems. The evidence set is claim-specific, and omission of a potentially relevant basis needs justification (Messick, 1995b, pp. 744, 747). Using the aggregate as a predictive index needs target-matched out-of-sample performance. A causal explanation needs process evidence and interventions against plausible rival routes where access and the study design permit them. Removing a component and observing an effect establishes dependence over the tested levels; it doesn't by itself identify an internal representation or a corrective mechanism. A decision rule also needs defensible value scales, distributional tradeoffs, feasible alternatives, and decision performance.

Some accounts impose a stronger condition before calling an index a measure. Borsboom et al. (2004) require variation in the measured attribute to cause variation in the score. Within its tested range, held-out prediction may warrant use of a formula as a predictive index or decision input without showing that it measures a causally efficacious AGI attribute.

5.3 Measurement Interpretation: CHC and Comparability

CHC theory (Carroll, 1993; McGrew, 2009; Schneider & McGrew, 2018) can enter AI evaluation in four roles. A taxonomy organizes content. A factor structure records which task scores vary together in a declared system population. A predictive input forecasts external outcomes. A functional or causal interpretation explains dependence. Evidence for one role doesn't establish the next. The framework proposed by Hendrycks et al. (2025) principally supplies the first; the other claims require additional study. For the law firm, the difference is whether the developer's profile is a content inventory or evidence about abilities its practice areas would draw on. A taxonomy supports the first reading only.

A distinction from cognitive measurement clarifies why these roles have to remain separate. Embretson (1983, pp. 179–181, 195–196) distinguishes CONSTRUCT REPRESENTATION, the claim that specified processes, strategies, and knowledge stores underlie item responses, from NOMOTHETIC SPAN, the score's network of relations with measures, criteria, groups, and tasks. Either can be strong while the other is weak.

Projectibility is assessed separately for each of these claims (Section 2). A process model doesn't itself establish external reach, and predictive success doesn't show that the proposed process generated the score. For scores derived from factor models, too, validity is relative to a purpose and depends on analysis of both test and criterion tasks (Carroll, 1993, p. 693).

In factor analysis, a LOADING records how strongly a task score is associated with an inferred factor; a POSITIVE MANIFOLD is widespread positive correlation among task scores; and a GENERAL FACTOR is a single latent dimension introduced to account for much of that shared variation.

Task performances are observed, while broad and general abilities are inferred from their relations. Carrying human patterns of loadings or the positive manifold to artificial systems requires evidence in the target population. The rule for combining abilities is a separate choice: a target may permit additive compensation, impose minimum requirements, treat one domain as a bottleneck, or posit a causal chain in which one ability enables another.

Across 591 nominally distinct models and 12 tests, Ilić and Gignac (2024, pp. 4–5) find a positive manifold and a strong general factor. Their proposed hierarchy of four narrower factors beneath that general factor couldn't be interpreted; the narrower structure supported by their data combines knowledge with reading and writing. The largely verbal task set and the difficulty of identifying independent models also limit the result (Ilić & Gignac, 2024, pp. 7–9). The study supports a general factor in that model population while leaving projection of the human CHC decomposition to other system populations open.

Descriptive profiles can be compared without claiming that they measure identical latent abilities. Interpreting profile differences as differences in the same abilities across systems requires evidence that items and score relations retain the same interpretation: a linking or invariance argument. Depending on the design, comparability can be probed through rubric checks, family-specific item analyses, tests of whether particular items behave differently across system families after accounting for broader performance, and models that test whether item or factor relations remain comparable across groups (Meredith, 1993). Comparability is addressed assumption by assumption. Jung et al. (2026) give a direct illustration: across 17 language models, human psychometric instruments showed moderate reliability across item and prompt variations while their scores failed to align with, and sometimes ran opposite to, model behaviour on downstream tasks. Reliability and an interpretable score structure within a test don't establish external meaning for an artificial-system population.

Comparability isn't the only issue. The AGI label and the selection of domains also define which performances count as valuable. Content standards and score thresholds themselves need validation: higher score levels should reflect increasing complexity in the intended attribute rather than difficulty introduced by something outside it (Messick, 1995a, p. 6). Claiming that two systems meet the same standard presupposes comparability even when they aren't directly ranked (Messick, 1995a, p. 7). Once such a score informs action, the value choices become more explicit.

5.4 Decision Use

Decision framing specifies objectives, affected parties, consequences, feasible alternatives, preferences and tradeoffs, and hard constraints before a score is used to compare or optimize policies (Keeney & Raiffa, 1993). It also identifies who has authority to impose the resulting rule; evidence doesn't confer that authority, and a developer's source evidence doesn't transfer it. A numerical loss represents those preferences; it doesn't specify them. False reassurance, false alarms, severe tail events, and repeated exposure can have different costs.

WTL averages a selected tail of case risks or realized losses under a declared sampling measure; neither form automatically shows how consequences are distributed across people. When that distribution matters, index loss by the declared operator or affected-population stratum, and report stratum-conditional expected loss, tail loss, cumulative exposure, and applicable group constraints. Where the scored rubric is a surrogate, its link to realized consequences is a further projection that held-out task scoring doesn't establish; it needs exposure sampling and outcome surveillance in use.

Pooling across strata requires commensurable loss scales and a defensible rule for interpersonal or intergroup tradeoffs (Keeney & Raiffa, 1993). Calibration of a pooled quantity can't supply that rule. A hard WTL threshold prevents compensation by a good mean or profile but still permits tradeoffs within the selected tail; event-specific prohibitions or a maximum-loss constraint are needed when no such tradeoff is acceptable.

A dashboard can eliminate options that are no better on any relevant dimension without reducing the evidence to a scalar. Hard constraints, compensatory value or utility aggregation, and unconstrained predictive models used as inputs to a policy are other structures; they answer different questions. A release rule may combine a calibrated expected-loss model with a specified case-risk or realized-loss WTL threshold that the mean can't override, and it has to say whether that threshold applies jointly, by stratum, or to the maximum conditional tail; Section 4.2 gives a known-truth case in which the mean and the change tail are both stable while the case-risk tail is high. An imprecise estimate can't support a confident release, though it may strongly support a conservative one: restricted use, mandatory review, or no deployment.

Observed consequences prompt at least three questions. Do they reveal coverage of the intended attribute that is too narrow, score variation caused by something outside it, or a rival score interpretation? Do they undermine the relevance or utility claimed for this use? Do they make the policy unacceptable for independent distributive or ethical reasons even if the score interpretation and claimed utility survive? The answers can overlap, and favorable consequences can't rescue an invalid interpretation (Messick, 1995b, pp. 746–749). Accurate prediction of pooled loss likewise doesn't justify how a policy trades losses across affected parties.

Decision evidence is comparative. A proposed deployment rule should be assessed against feasible alternatives such as restricted use, mandatory review, a different workflow, or no deployment; comparison with alternatives also exposes its value commitments (Messick, 1995b, p. 748). Thresholds for monitoring, release, or independent review have to be calibrated against relevant outcomes under those alternatives.

6 Workflow and Limits

The workflow begins by registering the claim and fixing its bearer: base-model and deployed-system claims have to remain separate, and every transport step between them has to be identified.

Responsibility is divided but not dissolved. Developers are responsible for evidence supporting the interpretations and uses they advertise, while deployers and decision-makers remain responsible for the choices they make; foreseeable consequences create shared obligations (Kane, 2013, pp. 57–58). A developer can characterize a versioned model or reference configuration over declared task and perturbation ranges, estimate selected paired-condition quantities with uncertainty, test component dependencies, and publish the source evaluation design, population, and unsupported facets. Those findings remain conditional on the developer's sampling measure and declared loss.

A frontier developer can't anticipate every downstream task mix, workflow, operator population, affected population, consequence scale, or alternative policy. That practical limit narrows the claims its evidence can support; it doesn't make local validation unnecessary. The deployer completes the declaration for the full bearer, supplies application-specific relevance and utility evidence, and tests the remaining transport step using its own task or episode sampling frame (Messick, 1995b, p. 744).

Applied to the running law-firm example, the firm compares the workflow with feasible alternatives on held-out local matters, monitors intended and unintended consequences and their distribution in use, and reassesses after material change or drift. Evidence may combine developer and independent studies, justified sharing across sites, synthetic and retrospective local cases, shadow deployment (running alongside live work without controlling outcomes), prospective surveillance, and incident review. Thin evidence widens uncertainty and narrows permitted use.

6.1 Operational Sequence

Most behavioural and predictive parts of the framework can be implemented without model internals. Measurement and causal claims may require additional process or intervention evidence. The operational sequence is:

1. Register the claim and use. Complete the ten-field projection declaration in Section 2, including prospective support, defeat, and inconclusive conditions, and assign source-side and target-side responsibilities. An exploratory analysis completes the same fields but reserves fresh evidence for later confirmation.
2. Audit the sampling facets. Record systems and lineages, item and template clusters, domains, perturbation families and realizations, operator roles, teams or sites, affected-population and exposure strata, response replicates, scorers, and time; mark facets as crossed or nested and their levels as fixed or sampled.
3. Fix the measurement and decision design. Specify content and process rationales, rubrics, target-population weights, sampling budget, and the distinction between fixed-set, stochastic-response, and population inference. For an aggregate to be interpreted as a measure, state and justify the combination rule, whether compensatory, conjunctive, or bottleneck, and the score ranges over which any substitutions are claimed. For a decision, specify affected parties, feasible alternatives, losses and distributional tradeoffs, constraints, and thresholds. For a paired-condition design, also specify paired units, perturbations, controls, tail fraction, minimum count, and the candidate outputs implicated by the claim. Replace general stability language with the bounded senses being tested.
4. Collect the observations. For paired-condition designs, balance conditions, randomize order where possible, and retain item identifiers. In every design, prevent data used for adaptation, selection, or threshold setting from leaking into final estimation.
5. Publish description and risk separately. For paired conditions, report the preregistered behavi-

oural outputs and their uncertainty, publish the raw domain profiles and their spreads alongside r, retain the paired source data, and publish the full module when all of its bounded senses are in scope. Distinguish intervals for individual comparisons from intervals or error rates that account for all reported comparisons together. For nonlinear item and tail summaries, distinguish raw, null-referenced, split-sample, and cross-fitted estimates where used, and state which tail each of them targets. When a target-conditioned risk report is in scope, state whether its absolute tail is over conditional case risks or realized losses, report the condition levels as primary results and their difference as secondary, and give the loss scale, any surrogate–consequence link, joint target distribution, residual randomness, exposure assumptions, and conditional or noncompensatory constraints.

6. Test each source–target link. For a measurement interpretation, examine the content, process, scoring, and comparability bases implicated by the claim and justify any omitted basis. For prediction, use matching external evidence, hold out at the independent unit declared for the projection, limit the target-specific model to preregistered predictors, and report the dependence model, calibration, target loss, incremental value, and uncertainty. For a causal explanation, intervene where possible and test rivals; a component-removal effect establishes dependence over the tested levels, not by itself an internal representation or corrective mechanism. For a decision, compare feasible policies under the stated values, constraints, and distribution of consequences.
7. Report scope and limits. Report unsupported or unresolved links as such. A composite claim spanning two or more tested links has to be declared and argued for, not read off their conjunction. Composition isn't unavailable: links can chain when the target population of one is the source population of the next, the assumptions at that interface are compatible, dependence and effect modification are represented, no facet held fixed in one study varies uncontrolled in the next, and uncertainty is propagated along the chain. What fails is composition by accumulation, where a series of bounded successes is treated as warranting a claim none of them declared. Record any demotion to a fallback scope, the reach given up, and whether what remains still meets the declared minimum. Label additional granular predictors as exploratory unless they were needed for the declared description or shown to add preregistered value for the target.
8. Maintain a two-sided projection record. Publish a developer evidence card with the source design and population, code and data provenance, prompts, seeds, model and scorer versions, exclusions, missingness, compute use, split assignments, granular results, uncertainty, rubrics, known limits, and update triggers. Join it to a deployer addendum specifying the target design and population, bearer, operator and affected populations, exposure, local losses and constraints, alternatives, evidence sources, version mapping, local results and deviations, and monitoring. Reopen the argument after material system, population, workflow, or outcome change.

6.2 Limitations

The framework itself is untested against deployment outcomes. The reanalysis and simulations verify what the quantities distinguish, not that any of them predicts a target.

Repeated sampling and tail estimation can be expensive. Adaptive allocation may reduce cost but adds selection and dependence. Deterministic fixed-set reports avoid repetition cost while remaining silent about wider item or context populations. Group-conditional assessment may require sensitive data and enough exposure in each stratum; without them, the corresponding claim is limited and a pooled substitute isn't licensed.

The estimators carry their own limits. Null-referenced contrasts are diagnostics against a baseline-only simulation rather than general bias corrections, and split-sample or cross-fitted tail estimates target the selected set rather than the population tail.

A perturbation is informative only relative to a property expected to remain invariant. Task-irrelevant context is designed to preserve the answer; distribution shifts, tool changes, and component removal may change the task itself. The evaluator has to defend the comparison.

Tail membership and item effects may not transfer across models or versions. Zhang et al. (2026, pp. 2, 4–6) report a mean cross-model correlation of .00 for per-item change and a mean Jaccard overlap of .09 for the most-affected decile on MMLU-Pro, making a common risk set across models or versions an empirical hypothesis.

7 Conclusion

An aggregate can describe its construction sample without supporting a broader inference. Generalizing from those observations to a declared evaluation universe and extrapolating from that universe to other tasks, contexts, systems, or times are distinct inferential steps.

The paired-condition module distinguishes profile shape, signed level, two-sided item change and its directional components, worst-tail degradation, case-risk tails, and realized-loss tails. None is a universal proxy for the others: the declared target identifies distinctions that may need to be preserved, while matched evidence determines which earn a place in a target-specific report. An aggregate interpreted as a measure or used in a decision also has to earn a common scale and acceptable tradeoffs among its components.

The reanalysis shows a small average decline coexisting with a much larger change tail, and the simulations show a change tail of zero coexisting with a high case-risk tail. Both show that raw, null-referenced, and split-sample estimators target different objects. Neither shows that any quantity predicts an untested target.

Goodman's riddle shows why fit to source observations can't, by itself, determine which continuation is supported. Validity belongs to a proposed interpretation and use; within that argument, projectibility is the evidential standing of a source–target link that extends an interpretation, prediction, or explanation beyond the observed responses. The practical discipline is to declare the link's source, target, bearer, populations, intervention range, horizon, risk measure, decision criterion, tolerance, evidential test, and failure conditions; audit the facets that generated the observations; retain potentially relevant resolution; and assemble evidence appropriate to that relation.

A CHC taxonomy may organize benchmark content, but claims that AI scores exhibit its narrower factor structure, remain comparable across systems, support external prediction, or provide evidence of AGI require separate support. Measurement interpretations need a defensible account of the evidential bases they implicate; predictions need target-matched external tests; and causal explanations need process evidence and tests against rivals. Decisions also require defensible values, alternatives, institutional authority, and continuing scrutiny of consequences.

Developers should characterize source evaluations and expose unsupported facets; deployers should complete local warrants, monitor whether the target conditions continue to obtain, and monitor consequences in use. What the law firm has to warrant is its own release argument; the developer's multidomain scores and the generality claim attached to them aren't that argument, and no accumulation of them becomes it. Open-ended AGI claims require more than accumulated coverage. Warrant may combine bounded empirical links with substantive or formal rationale for extending them, but successful links don't automatically compose into unrestricted generality. A warranted inference from an aggregate rests on a declared, revisable argument whose evidence is matched to the scope of each claim.

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